HARBORTECH SOLUTIONS

Week 6 | Workload-Fit Matrix | TKT-2026-0006

Client: Bright Path Community Services

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1. WORKLOAD-FIT MATRIX: A, B, AND C

Workload A: Serverless (AWS Lambda, triggered by EventBridge on a daily schedule). The evidence is that it runs once each morning, finishes quickly, keeps no local state between runs, and sits idle otherwise. Short, scheduled, stateless, bursty work is what serverless is built for, and you pay only for the seconds it runs. A VM or always-on container would sit idle and incur cost for most of the day.

2. LAMBDA, API GATEWAY, AND REMAINING RESPONSIBILITIES

Workload A: Lambda alone is enough, since an EventBridge schedule triggers it and there are no inbound HTTP requests. API Gateway isn't needed.
Workload B: Both. API Gateway provides the REST endpoint and handles request routing, throttling, and authorization. Lambda runs the code behind it.
What AWS takes over: Provisioning, patching, and scaling servers, managing the operating system and runtime infrastructure, and keeping capacity available.
What remains with the customer:
Function code and its dependencies.
IAM permissions with least privilege.
Secrets and configuration.
Data protection and API authentication and authorization.
Timeout and memory settings.
Logging, monitoring, and alerting.
Cost controls and throttling limits.
Serverless reduces server management but doesn't remove responsibility for the application and its security.

3. VALIDATION ASSUMPTION AND INTERN ESCALATION

Assumption to validate: Confirm that Workload A's run really is stateless and finishes within Lambda's limits (the 15-minute maximum timeout, memory, and any temporary storage needs). Also confirm that Workload B's traffic stays intermittent. If B's traffic is steady or high, or its latency needs are strict, containers could be cheaper or more predictable than Lambda.
Decision for the intern to escalate: Any decision that affects security, cost, or availability. For example, which IAM permissions the function gets, whether the API is public or authenticated, or whether to move the custom-OS Workload C off its VM. The intern can implement the approved design but shouldn't unilaterally make these calls.